

Scoping review protocol

Title

Reframing the malaria-sickle cell disease relationship in sub-Saharan Africa as a bounded syndemic: A scoping review and evidence gap map protocol

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1. Protocol information

Review type: Scoping review

Methodological approach: JBI-informed scoping review methodology

Reporting guideline: PRISMA Extension for Scoping Reviews, PRISMA-ScR

Planned registration: Open Science Framework or another suitable open protocol registry

Protocol version: Version 1.0

Protocol date: 10/04/2026

Review status at registration: Searches not completed, screening not started

Geographic focus: Sub-Saharan Africa

Theoretical orientation: Bounded syndemic framework

This protocol follows JBI guidance for scoping reviews, which recommends clearly specifying the review question, inclusion criteria, search strategy, source selection, data extraction, and evidence presentation. JBI also recommends using the Population, Concept, Context framework to structure scoping review eligibility criteria (1). The final review will be reported using PRISMA-ScR, which was developed to improve transparency and reporting quality in scoping reviews (2).

2. Background and rationale

Malaria and sickle cell disease (SCD) are major and intersecting health problems in sub-Saharan Africa (SSA). Malaria remains highly concentrated in the African region, with *Plasmodium falciparum* responsible for most severe malaria morbidity and mortality. Continental mapping studies show that malaria burden has declined in some settings because of control interventions, while remaining spatially heterogeneous and persistently concentrated in many parts of SSA (3, 4). SCD is also heavily concentrated in SSA and has been described as a neglected contributor to early childhood mortality, partly because many affected children die before diagnosis or before sustained contact with organised care (5). The Global Burden of Disease 2021 SCD analysis further shows that cause-specific mortality underestimates the total mortality attributable to SCD because SCD contributes to deaths assigned to proximate causes such as infection, severe anaemia, respiratory disease, stroke, and other complications (6).

The biological relationship between malaria and SCD is grounded in the protective effect of sickle cell trait (HbAS) against severe *P. falciparum* malaria. Cohort evidence from Kenya shows that HbAS is associated with protection against severe malaria and malaria-related childhood outcomes (7). This protective effect has contributed to the persistence of haemoglobin S (HbS) in malaria-endemic populations. However, homozygous inheritance of HbS produces sickle cell anaemia (SCA), a severe inherited haemoglobinopathy associated with chronic haemolysis, baseline anaemia, acute complications, infection vulnerability, and early mortality in the absence of comprehensive care. Geostatistical mapping has shown broad geographic correspondence between HbS frequency and historical malaria endemicity, especially in Africa,

and has demonstrated marked spatial heterogeneity in HbS frequency and estimated SCA births (8, 9).

The malaria and SCD relationship is not limited to HbS. Malaria has shaped a wider landscape of human red blood cell variation. Haemoglobin C, alpha-thalassaemia, glucose-6-phosphate dehydrogenase deficiency, ABO blood group, Dantu, and other erythrocyte variants have been associated with malaria susceptibility or severity through mechanisms such as altered parasite invasion, reduced rosetting, impaired cytoadherence, erythrocyte membrane changes, and phenotype-specific protection (10-13). This broader red blood cell polymorphism literature is relevant because malaria-endemic African populations may carry multiple variants, and their effects may be phenotype-specific or modified by co-inheritance.

Clinically, malaria and SCD converge most clearly through severe anaemia, acute febrile illness, malaria-positive presentations, respiratory distress, transfusion need, and diagnostic attribution. Malaria contributes to anaemia through destruction of infected and uninfected erythrocytes, inflammation, impaired erythropoiesis, and splenic clearance, while SCA produces chronic haemolysis, reduced red cell survival, baseline anaemia, and reduced physiological reserve (14, 15). The Transfusion and Treatment of African Children Trial established severe anaemia as a major paediatric hospital syndrome in Uganda and Malawi, with transfusion timing and volume affecting outcomes in defined groups of children (16, 17).

Evidence from a secondary analysis of the TRACT trial directly illustrates the under-recognition problem. Among Ugandan children admitted with severe anaemia, 1,038 of 3,483 had SCA, and 608 had previously unknown SCA identified through post-trial genotyping rather than routine clinical recognition. Among malaria-positive children, those with SCA had lower parasite densities and lower *P. falciparum* histidine-rich protein 2 concentrations than children with normal haemoglobin genotype, despite severe anaemia (18). This suggests that children with SCA may present with malaria-positive illness and severe anaemia while the underlying haemoglobinopathy remains hidden within routine clinical categories.

Health-system conditions shape whether SCD burden is recognised, treated, recorded, or missed. Early diagnosis, newborn screening, confirmatory testing, caregiver counselling, infection prevention, vaccination, penicillin prophylaxis, hydroxyurea, transfusion readiness, referral, and

long-term follow-up are essential elements of SCD care. WHO identifies early diagnosis, newborn screening, vaccination, hydroxyurea, blood transfusion, and comprehensive management as important elements in SCD prevention and care (19). Penicillin prophylaxis has strong evidence for reducing pneumococcal septicemia among young children with SCA (20). Hydroxyurea has been shown to be feasible and beneficial for children with SCA in SSA, with reductions in pain, infections, malaria episodes, transfusions, and death (21).

Newborn screening programmes and implementation networks have demonstrated that early diagnosis is feasible in African settings, while also showing that screening must be linked to confirmation and long-term care. Ghana's newborn screening programme showed that large-scale infant screening and active follow-up could be implemented in an African setting (22). Uganda's sickle surveillance study showed that dried blood spot platforms can reveal national and subnational sickle-related burden (23). The Consortium on Newborn Screening in Africa has advanced a multicountry implementation model linking newborn screening to early standardised care and mortality ascertainment (24). Recent newborn screening reviews emphasise that screening benefit depends on result return, confirmatory diagnosis, caregiver notification, counselling, clinic enrolment, preventive care, and long-term follow-up (25, 26).

This review is framed by syndemic theory. Syndemic theory examines how health conditions cluster, interact, and produce excess burden within harmful social, ecological, or structural conditions (27, 28). However, methodological critiques warn that syndemic claims should not be inferred from co-occurrence alone. They should specify the type of relationship being claimed, such as biological interaction, disease clustering, structural amplification, or excess burden beyond additive effects (29-32). For this review, malaria and SCD will be conceptualised as a bounded, serially causal, structurally mediated syndemic pathway. The review will not assume that all malaria and SCD co-occurrence is syndemic interaction. Instead, it will map whether the evidence supports co-occurrence, spatial co-risk, biological interaction, clinical convergence, service-mediated visibility, structural vulnerability, or programme response.

A scoping review is appropriate because the evidence base is heterogeneous and distributed across multiple disciplines, including malaria epidemiology, human genetics, paediatric haematology, spatial epidemiology, health systems, newborn screening, implementation science, routine data, and syndemic theory. Scoping reviews are intended to map evidence, clarify

concepts, identify evidence types, and locate knowledge gaps rather than estimate a pooled intervention effect (1, 2). This review will therefore map and configure the evidence base.

3. Review aim

The aim of this scoping review is to map and synthesize evidence on the malaria and SCD relationship in SSA using a bounded syndemic framework, with attention to biological interaction, spatial co-risk, clinical convergence, service-mediated visibility, structural vulnerability, programme response, and use of syndemic theory.

4. Review objectives

The review will address the following objectives:

1. To map evidence on biological interaction between *P. falciparum* malaria and HbS or other malaria-relevant red blood cell polymorphisms in SSA.
2. To describe evidence on spatial co-risk between malaria exposure, HbS distribution, SCD prevalence, and SCA birth burden.
3. To synthesize evidence on clinical convergence between malaria and SCD, especially severe anaemia, malaria-positive illness, acute febrile illness, respiratory distress, transfusion need, and diagnostic attribution.
4. To map evidence on under-recognition, diagnostic misclassification, and hidden SCD burden in paediatric malaria, severe anaemia, hospital admission, routine data, and mortality data.
5. To describe evidence on service-mediated visibility, including newborn screening, confirmatory diagnosis, admission, coding, reporting, clinic enrolment, referral, and mortality attribution.
6. To synthesize evidence on structural vulnerability and access to essential SCD services, including prophylaxis, vaccination, hydroxyurea, transfusion readiness, specialist care, referral, retention, and follow-up.

7. To describe programme responses, including newborn screening, point-of-care testing, early standardised care, screening-to-care linkage, decentralised care, referral systems, and implementation networks.
8. To assess how syndemic theory has been explicitly used, implicitly reflected, or omitted in malaria and SCD research in SSA.
9. To configure mapped evidence into a bounded malaria-SCD syndemic model for future research and programme planning.

5. Review questions

5.1 Main review question

How has the literature conceptualised, measured, and linked malaria-SCD biological interaction, spatial co-risk, clinical convergence, service-mediated visibility, structural vulnerability, and screening-to-care systems in sub-Saharan Africa?

5.2 Sub-questions

1. What evidence exists on the biological interaction between *P. falciparum* malaria and HbS or other malaria-relevant red blood cell polymorphisms?
2. How has the spatial relationship between malaria exposure, HbS distribution, SCD prevalence, and SCA birth burden been mapped or modelled?
3. What evidence exists on clinical convergence between malaria and SCD, especially severe anaemia, malaria-positive illness, respiratory distress, transfusion need, acute febrile illness, and severe malaria phenotypes?
4. What evidence exists on under-recognition, diagnostic misclassification, or hidden SCD burden in paediatric malaria, severe anaemia, hospital admission, routine reporting, or mortality data?
5. How do routine health information systems, facility admissions, diagnosis, coding, referral, reporting, and data quality shape service-visible SCD burden?

6. What evidence exists on access to essential SCD services, including newborn screening, confirmatory diagnosis, counselling, prophylaxis, vaccination, hydroxyurea, transfusion readiness, referral, retention, and follow-up?
7. What evidence exists on programme response, including newborn screening, point-of-care testing, early standardised care, screening-to-care linkage, decentralised service delivery, and referral systems?
8. How has syndemic theory been used, implied, or omitted in malaria and SCD research in SSA?
9. What evidence gaps remain in integrating malaria-SCD biology, spatial co-risk, clinical under-recognition, routine visibility, structural vulnerability, and screening-to-care systems?

6. Conceptual framework

This review will use a bounded syndemic framework. The proposed pathway is:

1. Malaria ecology contributes to HbS selection and persistence.
2. HbS persistence produces continuing HbSS births and SCA burden.
3. SCA modifies vulnerability to severe anaemia, infection, acute febrile illness, respiratory distress, and physiological decompensation.
4. Malaria and SCA converge clinically through severe anaemia, malaria-positive illness, transfusion need, acute paediatric deterioration, and diagnostic attribution.
5. Health systems determine whether affected children are screened, confirmed, recognised, recorded, treated, referred, and retained in care.

6. Structural conditions, including geography, poverty, laboratory access, blood availability, medicines, workforce, referral capacity, and data systems, shape outcomes and visibility.
7. Programme response converts hidden or expected burden into diagnosis, care enrolment, prevention, treatment, referral, and follow-up.

The review will classify evidence into seven syndemic-relevant categories:

1. Co-occurrence
2. Spatial co-risk
3. Biological interaction
4. Clinical convergence
5. Service-mediated visibility
6. Structural vulnerability
7. Programme response

Co-occurrence alone will not be treated as sufficient evidence of a syndemic relationship. Stronger syndemic relevance will be assigned to studies that connect two or more categories, especially biological interaction, clinical convergence, service-mediated visibility, structural vulnerability, and programme response.

7. Eligibility criteria

Eligibility criteria will be structured using the Population, Concept, Context framework recommended for scoping reviews by JBI (1).

Table 1. Eligibility criteria using Population, Concept, Context (PCC)

PCC element	Inclusion criteria	Exclusion criteria	Rationale
Population	Human populations in SSA, including neonates, infants, children, adolescents, adults, pregnant women where relevant to SCD birth burden, individuals with SCD, individuals with SCA, individuals with HbSS, individuals with HbAS or sickle cell trait, and populations carrying malaria-relevant red blood cell polymorphisms.	Animal-only populations with no human relevance. Studies of populations outside SSA with no conceptual, methodological, or comparative relevance to SSA.	The review focuses on human malaria-SCD relationships in SSA, including genetic, clinical, epidemiological, and health-system dimensions.
Population, health-system actors	Caregivers, health workers, laboratory staff, programme implementers, health facilities, newborn screening programmes, routine health information systems, transfusion services, referral	Health-system studies with no relevance to malaria, SCD, HbS, severe anaemia, screening, routine visibility, or care systems.	The review examines how health systems make SCD burden visible or hidden and how programmes respond.

	networks, and SCD service systems where they address the review concepts.		
Concept, biological interaction	Malaria-HbS interaction, HbAS protection, HbS persistence, HbSS or SCA burden, red blood cell polymorphisms, HbC, alpha-thalassaemia, G6PD deficiency, ABO blood group, Dantu, epistasis, parasite invasion, cytoadherence, rosetting, severe malaria protection, and malaria phenotype modification.	Biological studies with no relevance to malaria, SCD, HbS, red blood cell polymorphisms, or human disease interpretation.	This concept captures the evolutionary and mechanistic foundation of the malaria-SCD relationship.
Concept, spatial co-risk	Malaria exposure, malaria endemicity, parasite prevalence, malaria incidence, HbS distribution, SCD prevalence, SCA birth burden,	Spatial studies unrelated to malaria, SCD, HbS, child health, screening, or care access.	The review examines whether malaria and inherited sickle-related risk overlap spatially and how such evidence informs planning.

	geospatial mapping, spatial heterogeneity, co-risk, co-clustering, facility catchments, geographic accessibility, and spatial prioritisation.		
Concept, clinical convergence	Severe anaemia, malaria-positive illness, severe malaria phenotypes, acute febrile illness, respiratory distress, transfusion need, acute paediatric deterioration, hospital admissions, diagnostic attribution, SCD under-recognition, or hidden SCA among clinical syndromes.	Clinical studies of malaria or SCD with no relevance to convergence, severe anaemia, diagnostic attribution, or under-recognition.	This concept captures how malaria and SCD meet in clinical practice.
Concept, service-mediated visibility	Newborn screening, confirmatory diagnosis, clinical recognition, admission, coding, routine reporting, DHIS2, HMIS, clinic enrolment, referral,	Routine data studies with no link to malaria, SCD, HbS, severe anaemia, diagnosis, coding, reporting, or mortality attribution.	This concept addresses how burden becomes visible or invisible in health systems.

	mortality attribution, routine surveillance, service-visible burden, hidden burden, or undercounting.		
Concept, structural vulnerability	Unequal access to screening, diagnosis, counselling, penicillin prophylaxis, vaccination, hydroxyurea, transfusion readiness, specialist care, referral, retention, follow-up, medicines, laboratory systems, workforce, transportation, and facility readiness.	General social determinant studies with no relevance to SCD, malaria, screening, care access, or programme response.	Structural vulnerability is central to understanding preventable morbidity, mortality, and invisibility.
Concept, programme response	Newborn screening, point-of-care testing, early standardised care, screening-to-care linkage, caregiver notification, counselling, clinic enrolment, decentralised care,	Programme reports that do not provide substantive information on SCD, malaria, screening, linkage to care, or service response.	The review maps how evidence has been translated into programme action.

	hub-and-spoke models, referral systems, implementation networks, and service strengthening.		
Concept, syndemic theory	Explicit use of syndemic theory, implicit syndemic logic, disease clustering, biological interaction, clinical convergence, structural amplification, excess burden, or programme response.	Papers using syndemic terminology without relevance to malaria, SCD, SSA, or the review framework.	The review evaluates whether syndemic theory is used explicitly or implied through evidence structure.
Context	SSA countries and malaria-endemic African health systems, including community, facility, laboratory, programme, district, national, multicountry, or regional settings.	Studies conducted wholly outside SSA unless they provide foundational conceptual or intervention evidence directly relevant to the review.	The review is designed to inform SSA-focused research and programme planning.
Evidence source type	Empirical studies, clinical studies, cohort studies, case-	Records without sufficient information for charting.	Scoping reviews are suited to heterogeneous

	control studies, cross-sectional studies, geospatial studies, modelling studies, implementation studies, programme evaluations, trials relevant to SCD care, routine data studies, mortality studies, reviews, policy documents, technical reports, and relevant grey literature.	Duplicate reports where a fuller report is available. Opinion pieces without substantive conceptual, empirical, policy, or programme content.	evidence sources where the goal is mapping and conceptual synthesis.
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8. Inclusion criteria

A source will be included if it meets all of the following criteria:

1. It addresses malaria, SCD, HbS, a malaria-relevant red blood cell polymorphism, severe anaemia, newborn screening, SCD care, service visibility, structural vulnerability, programme response, or syndemic theory in a way relevant to the review question.
2. It relates to SSA or to malaria-endemic African health systems.
3. It includes human, clinical, epidemiological, health-system, programme, policy, spatial, modelling, implementation, or theoretical relevance.
4. It provides sufficient information for charting.
5. It contributes evidence to at least one syndemic-relevant category: co-occurrence, spatial co-risk, biological interaction, clinical convergence, service-mediated visibility, structural vulnerability, or programme response.

9. Exclusion criteria

The review will exclude:

1. Studies outside SSA with no direct conceptual, methodological, or comparative relevance to SSA.
2. Malaria studies with no relevance to SCD, HbS, red blood cell polymorphisms, severe anaemia, clinical convergence, or syndemic interpretation.
3. SCD studies with no relevance to malaria-endemic settings, malaria exposure, under-recognition, screening, service access, structural vulnerability, or programme planning.
4. Animal-only studies with no direct relevance to human malaria-SCD relationships.
5. Laboratory-only studies without clinical, epidemiological, population genetic, mechanistic, or programme relevance.
6. Studies focused exclusively on non-SCD haemoglobinopathies unless they inform malaria-relevant red blood cell variation.
7. Records without sufficient information for extraction after full-text review.
8. Duplicate reports of the same data where a more complete or more recent source is available.
9. Editorials, opinion pieces, or commentaries without substantive conceptual, empirical, programme, or policy contribution.

10. Special eligibility decision rules

Issue	Decision rule
Studies outside SSA	Include only if they provide foundational or directly relevant conceptual, methodological, or intervention evidence. For example, a foundational trial of penicillin prophylaxis

	may be included because it informs essential SCD care pathways relevant to SSA.
Laboratory studies	Include only if they directly explain mechanisms relevant to malaria and SCD interpretation, such as HbAS protection, parasite invasion, cytoadherence, rosetting, red cell membrane biology, erythrocyte polymorphisms, or epistasis.
Malaria-only studies	Include only if they address HbS, red blood cell polymorphisms, severe anaemia, SCD-related vulnerability, malaria phenotype interpretation, or malaria-SCD programme relevance.
SCD-only studies	Include only if they address malaria-endemic settings, severe anaemia, under-recognition, newborn screening, routine visibility, access to care, structural vulnerability, or programme response.
Reviews	Include if they synthesize evidence relevant to the review question. Primary studies cited within included reviews may be screened separately if directly relevant.
Grey literature	Include if it provides substantive evidence on policy, programme design, newborn screening, service delivery, SCD burden, malaria burden, or health-system response.
Conference abstracts	Include only if they contain sufficient methods and findings for charting.

Duplicate or overlapping reports	Retain the most complete report. Retain secondary reports only if they contribute distinct domain-specific evidence.
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11. Types of evidence sources

Eligible sources will include peer-reviewed empirical studies, cohort studies, case-control studies, cross-sectional studies, hospital-based clinical studies, community surveys, birth cohort studies, newborn screening studies, geospatial studies, geostatistical modelling studies, mathematical or statistical modelling studies, routine data studies, mortality attribution studies, implementation studies, programme evaluations, clinical trials relevant to SCD care, systematic reviews, scoping reviews, narrative reviews with substantive synthesis, policy documents, technical reports, programme reports, relevant conference reports, and conceptual or theoretical papers.

12. Information sources

The following electronic databases will be searched:

1. PubMed or MEDLINE
2. Embase
3. Scopus
4. Web of Science Core Collection
5. Global Health
6. CINAHL
7. African Index Medicus

8. Cochrane Library

The following grey literature sources will be searched:

1. World Health Organization
2. Africa CDC
3. SickInAfrica
4. American Society of Hematology resources where relevant to CONSA
5. Ministries of Health in SSA countries
6. Global Burden of Disease resources
7. Newborn screening programme reports
8. Institutional repositories
9. Relevant conference proceedings
10. SCD advocacy and implementation networks
11. National SCD or non-communicable disease policy documents where available

Backward and forward citation searching will be conducted for key sources on malaria-HbS biology, HbS mapping, SCD burden, severe anaemia, newborn screening, hydroxyurea, transfusion, routine data, and syndemic theory.

13. Search strategy

The search strategy will be developed with an information specialist or medical librarian. Searches will combine controlled vocabulary and free-text terms adapted to each database.

13.1 SCD and HbS terms

“sickle cell disease” OR “sickle cell anaemia” OR “sickle cell anemia” OR SCD OR SCA OR HbSS OR HbAS OR “sickle cell trait” OR “haemoglobin S” OR “hemoglobin S” OR “Hb S” OR “sickle haemoglobin” OR “sickle hemoglobin”

13.2 Malaria terms

malaria OR “Plasmodium falciparum” OR falciparum OR parasitaemia OR parasitemia OR PfHRP2 OR “histidine-rich protein 2” OR “severe malaria” OR “cerebral malaria” OR “severe malarial anaemia” OR “severe malarial anemia”

13.3 Red blood cell polymorphism terms

“red blood cell polymorphism” OR “erythrocyte polymorphism” OR “alpha thalassaemia” OR “alpha thalassemia” OR “alpha-plus-thalassaemia” OR “alpha-plus-thalassemia” OR G6PD OR “glucose-6-phosphate dehydrogenase” OR “haemoglobin C” OR “hemoglobin C” OR HbC OR ABO OR “blood group O” OR Dantu OR epistasis

13.4 Clinical expression terms

“severe anaemia” OR “severe anemia” OR anaemia OR anemia OR transfusion OR “blood transfusion” OR “respiratory distress” OR “acute chest syndrome” OR fever OR infection OR “febrile illness” OR “malaria-like illness” OR “clinical phenotype” OR “severe illness” OR “paediatric admission” OR “pediatric admission”

13.5 Recognition and visibility terms

underdiagnosis OR under-recognition OR “missed diagnosis” OR “clinical recognition” OR misclassification OR attribution OR “routine data” OR “health information system” OR DHIS2 OR HMIS OR coding OR reporting OR “mortality attribution” OR “cause of death” OR “service-visible burden” OR surveillance

13.6 Screening and care terms

“newborn screening” OR “neonatal screening” OR “infant screening” OR “point-of-care testing” OR “confirmatory diagnosis” OR counselling OR counseling OR “caregiver notification” OR hydroxyurea OR hydroxycarbamide OR penicillin OR vaccination OR immunisation OR immunization OR transfusion OR referral OR retention OR “linkage to care” OR “continuum of care” OR “comprehensive care”

13.7 Syndemic and structural context terms

syndemic OR syndemics OR “structural vulnerability” OR “structural violence” OR “health system” OR “health systems” OR “service readiness” OR “spatial prioritisation” OR “spatial prioritization” OR “hub and spoke” OR “facility readiness” OR “access to care” OR “implementation science” OR “programme response” OR “program response”

13.8 Geography terms

“Africa South of the Sahara” OR “sub-Saharan Africa” OR SSA OR Africa OR names of individual SSA countries.

The final search strategy will combine:

1. SCD or HbS terms
2. Malaria or red blood cell polymorphism terms
3. SSA geography terms
4. Domain-specific terms where needed

The complete final search strategy for each database will be reported in Supplementary File 2.

14. Search limits

No date restrictions will be applied at the initial search stage unless database constraints require them. No language restrictions will be applied at the search stage where feasible. Records in

languages not readable by the review team will be translated where possible or assessed using abstracts and available translation support.

Human evidence will be prioritised. Animal-only studies will be excluded unless they provide mechanistic evidence directly relevant to human malaria-SCD interpretation and are judged essential by the biological expert panel.

15. Record management

Search results will be exported into reference management software. Duplicates will be removed using automated deduplication followed by manual checking. Records will then be uploaded into a screening platform.

The review team will maintain a search log documenting:

1. Database searched
2. Platform used
3. Date searched
4. Search string used
5. Limits applied
6. Number of records retrieved
7. Number of duplicates removed
8. Number advanced to screening

16. Study selection process

Study selection will occur in two stages.

16.1 Title and abstract screening

At least two reviewers will independently screen titles and abstracts against the eligibility criteria. Before formal screening begins, reviewers will pilot the screening form on a sample of records. Disagreements during pilot screening will be discussed to refine eligibility decision rules.

Each record will be coded as:

1. Include
2. Exclude
3. Unclear

Records coded as include or unclear by either reviewer will proceed to full-text screening.

16.2 Full-text screening

At least two reviewers will independently assess full texts for final inclusion. Reasons for exclusion will be recorded. Disagreements will be resolved through discussion. Unresolved disagreements will be adjudicated by the review lead or a third reviewer.

Full-text exclusion reasons will include:

1. Wrong geography
2. Wrong population
3. Wrong concept
4. No malaria relevance
5. No SCD or HbS relevance

6. No human, clinical, epidemiological, programme, or theoretical relevance
7. Insufficient information
8. Duplicate report
9. Laboratory-only without review relevance
10. Commentary without substantive contribution

The selection process will be reported using a PRISMA-ScR flow diagram.

17. Reviewer calibration

Before formal screening, reviewers will complete calibration exercises.

17.1 Title and abstract screening calibration

A pilot sample of records will be screened independently. The team will compare decisions and identify areas of disagreement. Eligibility criteria will be clarified where needed.

17.2 Full-text screening calibration

A sample of full-text articles will be screened independently. Reviewers will compare decisions and refine decision rules for ambiguous cases.

17.3 Data charting calibration

A sample of included studies will be charted independently by at least two reviewers. Extracted fields will be compared to ensure consistency in interpretation and coding. The data charting form will be revised where needed.

18. Data charting

A standardised data charting form will be developed in spreadsheet format. The form will be piloted and refined before full extraction.

18.1 Bibliographic information

1. Study ID
2. Author
3. Year
4. Title
5. Journal or source
6. Publication type
7. DOI or persistent identifier where available

18.2 Geographic information

1. Country
2. SSA region
3. Study location
4. Urban, rural, or mixed setting
5. Spatial scale, such as community, facility, district, region, national, multicountry, or continental

18.3 Study characteristics

1. Study aim

2. Study design
3. Data source
4. Sampling frame
5. Sample size
6. Population group
7. Age group
8. Inclusion criteria used by original study
9. Comparator group, where relevant

18.4 Disease and genetic variables

1. SCD status
2. SCA or HbSS status
3. HbAS or sickle cell trait status
4. HbS frequency
5. Other red blood cell polymorphisms assessed
6. Method of genotype or phenotype ascertainment
7. Confirmatory testing method

18.5 Malaria variables

1. Malaria exposure measure
2. Malaria infection measure
3. Parasite density
4. PfHRP2 or other parasite biomass marker
5. Malaria phenotype
6. Severe malaria definition
7. Malaria outcome
8. Malaria diagnostic method

18.6 Clinical variables

1. Severe anaemia
2. Haemoglobin threshold
3. Fever or acute febrile illness
4. Respiratory distress
5. Severe malaria features
6. Transfusion need

7. Admission outcome
8. Mortality
9. Diagnostic attribution
10. Evidence of SCD under-recognition

18.7 Health-system and visibility variables

1. Newborn screening
2. Confirmatory diagnosis
3. Result return
4. Caregiver notification
5. Clinic enrolment
6. Coding or reporting system
7. Routine health information system
8. Mortality attribution
9. Referral pathway
10. Follow-up and retention

11. Evidence of service-visible burden

12. Evidence of hidden or undercounted burden

18.8 Structural and programme variables

1. Penicillin prophylaxis

2. Vaccination

3. Hydroxyurea

4. Transfusion readiness

5. Specialist care

6. Facility readiness

7. Geographic access

8. Referral network

9. Decentralised care

10. Hub-and-spoke model

11. Programme response

12. Implementation barriers

13. Implementation facilitators

18.9 Syndemic coding variables

1. Co-occurrence
2. Spatial co-risk
3. Biological interaction
4. Clinical convergence
5. Service-mediated visibility
6. Structural vulnerability
7. Programme response
8. Explicit syndemic theory
9. Implicit syndemic logic
10. Type of relationship supported
11. Level of evidence integration

19. Evidence classification framework

Each included source will be coded into one or more syndemic evidence categories.

Category	Definition	Example evidence
Co-occurrence	Malaria and SCD, HbS, or SCA occur in the same	Malaria and SCD both described in SSA child populations.

	population, region, country, or clinical setting.	
Spatial co-risk	Malaria exposure, HbS frequency, SCD burden, SCA births, or child population need are geographically patterned, overlapping, or co-clustered.	Malaria and HbS maps, SCA birth-burden maps, subnational screening data.
Biological interaction	Malaria and red blood cell polymorphisms interact biologically.	HbAS protection, HbC, alpha-thalassaemia, G6PD, ABO, Dantu, epistasis, parasite invasion, rosetting.
Clinical convergence	Malaria and SCD converge clinically through severe anaemia, malaria-positive illness, transfusion need, acute febrile illness, respiratory distress, severe malaria phenotypes, or diagnostic attribution.	SCA among children admitted with severe anaemia, genotype-specific malaria biomarkers.
Service-mediated visibility	Screening, confirmatory diagnosis, clinical recognition, admission, coding, routine reporting, referral, clinic enrolment, or mortality attribution shape recorded SCD burden.	Previously unknown SCA found through genotyping, SCD-attributable mortality exceeding cause-specific SCD mortality.
Structural vulnerability	Unequal access to newborn screening, diagnosis, prophylaxis, vaccination,	High mortality where comprehensive care is incomplete, limited

	hydroxyurea, transfusion, specialist care, referral, or follow-up modifies outcomes, recognition, or survival.	hydroxyurea or transfusion access.
Programme response	Newborn screening, point-of-care testing, early standardised care, screening-to-care linkage, decentralised care, referral systems, service strengthening, spatial prioritisation, or implementation networks.	Ghana newborn screening, CONSA, SickleInAfrica, point-of-care testing studies.
Explicit syndemic theory	Direct use of syndemic theory, syndemic terminology, or formal syndemic constructs.	Studies explicitly framing malaria-SCD as a syndemic.
Implicit syndemic logic	Studies that do not use syndemic terminology but link disease clustering, biological interaction, clinical convergence, structural conditions, or programme response.	Studies connecting SCA under-recognition, severe anaemia, and service access without using syndemic terminology.

20. Level of evidence integration

Each included source will be coded by level of integration.

Level	Description	Example
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Level 1: Single-domain evidence	The study addresses one major domain.	HbAS protection, malaria mapping, hydroxyurea trial, newborn screening report.
Level 2: Two-domain evidence	The study links two domains.	HbS mapping and malaria endemicity, severe anaemia and SCA genotype.
Level 3: Multi-domain evidence	The study links three or more domains.	Spatial risk, severe anaemia, routine recognition, and service access.
Level 4: Syndemic-integrated evidence	The study explicitly or implicitly links biological interaction, spatial co-risk, clinical convergence, service-mediated visibility, structural vulnerability, and programme response.	Integrated malaria-SCD planning model using risk, visibility, service readiness, and screening-to-care evidence.

21. Data synthesis plan

Data synthesis will proceed in four stages.

21.1 Descriptive numerical summary

The review will summarise included evidence by:

1. Publication year
2. Country
3. SSA region
4. Study design

5. Population
6. Age group
7. Data source
8. Disease focus
9. Polymorphism assessed
10. Malaria outcome
11. SCD outcome
12. Evidence domain
13. Level of integration

21.2 Domain-based evidence mapping

Evidence will be organised into seven domains:

1. Biological interaction between malaria and red blood cell polymorphisms
2. Spatial co-risk and malaria-HbS or SCD geography
3. Clinical convergence and under-recognition
4. Service-mediated visibility and routine data
5. Structural vulnerability and access to essential SCD services

6. Programme response and screening-to-care systems

7. Use of syndemic theory in malaria-SCD literature

21.3 Thematic synthesis

The review will identify recurring themes within each domain, including HbAS protection, polygenic red blood cell adaptation, spatial heterogeneity of malaria and HbS, severe anaemia as clinical convergence, malaria positivity versus malaria-attributable illness, SCA under-recognition, routine data as service-visible burden, newborn screening as an entry point to care, structural gaps in hydroxyurea and transfusion access, screening-to-care cascade interruptions, and limited explicit use of syndemic theory.

21.4 Configurative synthesis

The review will integrate evidence across domains into a bounded malaria-SCD syndemic framework. The configurative synthesis will examine how evidence links malaria ecology, HbS selection and persistence, HbSS births and SCA burden, severe anaemia and clinical convergence, service-mediated recognition or under-recognition, structural access to essential care, and screening-to-care programme response.

22. Quality appraisal

Formal methodological quality appraisal will not be used as a criterion for inclusion. This is consistent with the purpose of a scoping review, which is to map the range, nature, and integration of evidence rather than to estimate pooled effects or exclude studies based on methodological quality.

However, methodological characteristics will be charted to support interpretation. These will include study design, sampling frame, population, data source, outcome definitions, genotype ascertainment method, malaria diagnostic method, spatial scale, and reporting limitations.

23. Handling of duplicates and overlapping data

When multiple publications report on the same dataset, programme, or cohort, the most complete or most relevant report will be prioritised for data charting. Secondary reports may be retained if they provide distinct findings relevant to another domain.

Overlapping evidence will be documented in the data charting form.

24. Handling of missing or unclear information

If information required for charting is missing or unclear, reviewers will record it as “not reported” or “unclear.” Authors may be contacted where missing information is essential for classification. If author contact is not feasible or unsuccessful, the study will be retained if enough information exists to support at least one evidence category.

25. Ethics

Ethical approval is not required because the review will synthesize published and publicly available evidence. No individual patient-level data will be collected by the review team.

26. Protocol registration

The protocol will be registered before screening begins, preferably on the Open Science Framework. The registration record will include the review title, aim, questions, eligibility criteria, information sources, search strategy summary, study selection approach, data charting approach, evidence classification framework, planned synthesis, and review team roles.

27. Protocol amendments

Any changes made after registration will be documented. The amendment log will include the date of amendment, section changed, description of change, reason for change, stage of review when change occurred, and person approving the change.

28. Data management and storage

Review records, screening decisions, data charting files, search logs, and supplementary tables will be stored in a secure shared project folder accessible to authorised review team members.

Version control will be used for the protocol, search strategy, screening guide, and extraction form.

The final dataset of included studies and data charting fields may be made available as supplementary material, subject to copyright and journal requirements.

29. Reporting plan

The final review will be reported using PRISMA-ScR. The manuscript will include a structured abstract, introduction, conceptual framework, methods, results by domain, cross-domain evidence configuration, discussion, strengths and limitations, and conclusion.

Supplementary files will include the protocol, full search strategies, screening guide, PRISMA-ScR checklist, data charting form, evidence classification codebook, characteristics of included studies, excluded full texts with reasons, domain-specific evidence tables, search results summary, reviewer calibration log, grey literature search log, conceptual model development notes, and evidence gap matrix.

30. Expected contribution

This scoping review is expected to contribute in three ways.

First, it will map the range of evidence linking malaria and SCD in SSA across biological, spatial, clinical, health-system, structural, programme, and theoretical domains.

Second, it will clarify the level of syndemic evidence available, distinguishing co-occurrence from spatial co-risk, biological interaction, clinical convergence, service-mediated visibility, structural vulnerability, and programme response.

Third, it will identify integrative gaps that limit research and programme planning, especially the lack of studies connecting malaria-SCD co-risk, hidden burden, service readiness, geographic access, and screening-to-care cascade outcomes.

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